(iii) $1325^\circ = (3 \times 360) + 180^\circ + 65^\circ$ the terminal side lies in III quadrant.

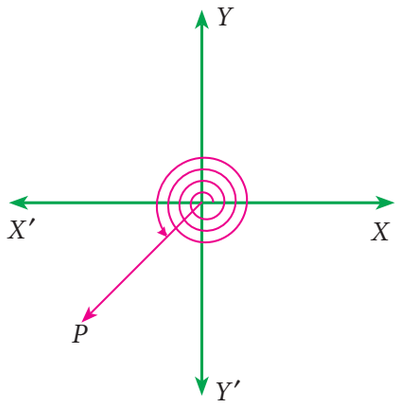
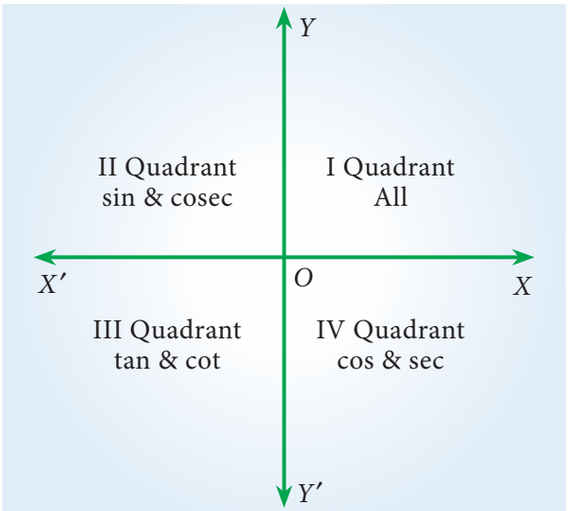


Fig. 4.6

4.1.2 Signs of the trigonometric ratios of an angle θ as it varies from 0° to 360°



ASTC : All Sin Tan Cos

Fig. 4.7

In the first quadrant both x and y are positive. So all trigonometric ratios are positive. In the second quadrant $(90^\circ < \theta < 180^\circ)$ x is negative and y is positive. So trigonometric ratios sin θ and cosec θ are positive. In the third quadrant $(180^\circ < \theta < 270^\circ)$ both x and y are negative. So trigonometric ratios tan θ and cot θ are positive. In the fourth quadrant $(270^\circ < \theta < 360^\circ)$ x is positive and y is negative. So trigonometric ratios cos θ and sec θ are positive.

A function $f(x)$ is said to be odd function if $f(-x) = -f(x)$. $\sin\theta$, $\tan\theta$, $\cot\theta$ and $\operatorname{cosec}\theta$ are all odd function. A function $f(x)$ is said to be even function if $f(-x) = f(x)$. $\cos\theta$ and $\sec\theta$ are even function.

4.1.3 Trigonometric ratios of allied angles:

Two angles are said to be allied angles when their sum or difference is either zero or a multiple of 90°. The angles $-\theta$, $90^\circ \pm \theta$, $180^\circ \pm \theta$, $360^\circ \pm \theta$ etc .,are angles allied to the angle θ. Using trigonometric ratios of the allied angles we can find the trigonometric ratios of any angle.

Angle/ Function	$-\theta$	$90^\circ - \theta$ or $\frac{\pi}{2} - \theta$	$90^\circ + \theta$ or $\frac{\pi}{2} + \theta$	$180^\circ - \theta$ or $\pi - \theta$	$180^\circ + \theta$ or $\pi + \theta$	$270^\circ - \theta$ or $\frac{3\pi}{2} - \theta$	$270^\circ + \theta$ or $\frac{3\pi}{2} + \theta$	$360^\circ - \theta$ or $2\pi - \theta$	$360^\circ + \theta$ or $2\pi + \theta$
Sine	$-\sin\theta$	$\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$-\sin\theta$	$\sin\theta$
Cosine	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$-\sin\theta$	$\sin\theta$	$\cos\theta$	$\cos\theta$
tangent	$-\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$	$\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$	$\tan\theta$
cotangent	$-\cot\theta$	$\tan\theta$	$-\tan\theta$	$-\cot\theta$	$\cot\theta$	$\tan\theta$	$-\tan\theta$	$-\cot\theta$	$\cot\theta$
secant	$\sec\theta$	$\operatorname{cosec}\theta$	$-\operatorname{cosec}\theta$	$-\sec\theta$	$-\sec\theta$	$-\operatorname{cosec}\theta$	$\operatorname{cosec}\theta$	$\sec\theta$	$\sec\theta$
cosecant	$-\operatorname{cosec}\theta$	$\sec\theta$	$\sec\theta$	$\operatorname{cosec}\theta$	$-\operatorname{cosec}\theta$	$-\sec\theta$	$-\sec\theta$	$-\operatorname{cosec}\theta$	$\operatorname{cosec}\theta$

Table : 4.1

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